

Suppose, you want to classify the three classes (A, B and C) from Table 1 dataset using three features: Color (Categorical), Size (Categorical) and Weight (Numeric).

| SL | Color | Size  | Weight | Class |
|----|-------|-------|--------|-------|
| 1  | Red   | Small | 1.2    | A     |
| 2  | Blue  | Large | 1.4    | B     |
| 3  | Green | Small | 1.4    | C     |
| 4  | Red   | Large | 1.5    | A     |
| 5  | Blue  | Small | 1.6    | C     |
| 6  | Red   | Small | 1.7    | B     |
| 7  | Green | Large | 1.7    | A     |
| 8  | Blue  | Large | 1.8    | C     |
| 9  | Green | Small | 1.9    | A     |
| 10 | Red   | Large | 2.0    | B     |
| 11 | Blue  | Small | 2.0    | C     |
| 12 | Red   | Small | 2.1    | A     |
| 13 | Green | Large | 2.1    | B     |
| 14 | Blue  | Large | 2.3    | C     |
| 15 | Red   | Small | 2.4    | A     |
| 16 | Green | Small | 2.4    | B     |

|    |       |       |     |   |
|----|-------|-------|-----|---|
| 17 | Blue  | Large | 2.5 | C |
| 18 | Green | Large | 2.6 | A |
| 19 | Red   | Small | 2.6 | B |
| 20 | Blue  | Large | 2.7 | C |

**Table 1**

### **Problem 1**

- a) Using GINI Index, construct the Classification Tree for Table 1. [Use Max\_Depth = 2]
- b) Using Information Gain, construct the Classification Tree for Table 1. [Use Max\_Depth = 2]

### **Problem 2**

You want to apply the Random Forest algorithm over Table 1. You are considering 3 estimators for the following three subsets where each subset consists of 10 instances:

| Estimator No. | Selected Instances      | Selected Features |
|---------------|-------------------------|-------------------|
| 1             | SL: 1 to 10             | Color, Weight     |
| 2             | SL: 11 to 20            | Size, Weight      |
| 3             | SL: 1 to 5 and 16 to 20 | Color, Size       |

Consider for each estimator, Max\_Depth = 2 and criterion='gini'. Now, suppose you have a new instance for testing:

[Blue, Small, 2.6]

Using Bagging technique, what would be the predicted class for this instance?

### **Problem 3**

Using the same data in Table 1, you want to create a classifier named “A vs Not A”. So, you have to consider class B & C instances as “Not A”. You want to apply the AdaBoost Algorithm now. Generate the first stump (use criterion='gini') only, calculate the amount of say and show the updated parameters.